Condensation and Metastability in the supercritical two-species zero-range process via resolvent and H^1-approximation

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Joint work with Seonwoo Kim

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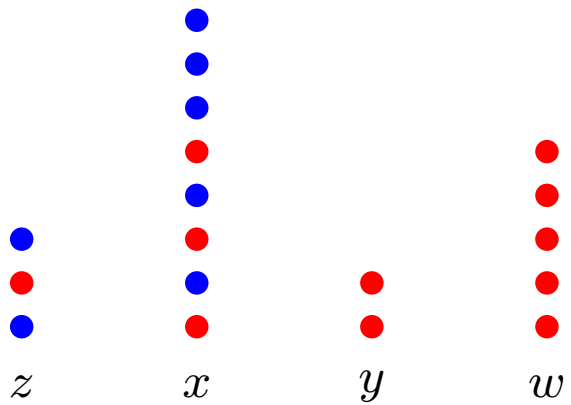
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Two-species zero-range process

- **Two random walks on finite space S**
  - ⋆ Irreducible jump rates $r_1, r_2 : S \times S \to [0, \infty)$.
  - ⋆ Normalized stationary measures m_1, m_2, i.e, $\max_{x \in S} m_i(x) = 1$.
  - ⋆ $S_\star = \{x \in S : m_1(x) = m_2(x) = 1\}$
- **Configuration space** $\Omega_N = \{\boldsymbol{\eta} = (\eta_1, \eta_2) \in \mathbb{N}_0^S \times \mathbb{N}_0^S : |\eta_1| = A,\ |\eta_2| = B\}$
  - ⋆ $\boldsymbol{\eta} = (\boldsymbol{\eta}(x))_{x \in S},\ \boldsymbol{\eta}(x) = (\eta_1(x), \eta_2(x)) \in \{0, 1, \dots, A\} \times \{0, 1, \dots, B\}$
  - ⋆ $\rho_N := A/N \to \rho \in (0, 1)$
- **Dynamics** Species i particle jumps $x \to y$ at rate $\boldsymbol{g}_i(\boldsymbol{\eta}(x)) r_i(x, y)$

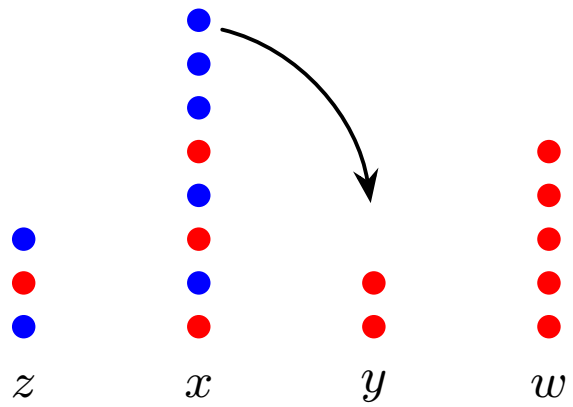
$$\boldsymbol{g}_i(k_1, k_2) = \begin{cases} \frac{k_i}{k_1+k_2} \frac{\mathfrak{a}(k_1+k_2)}{\mathfrak{a}(k_1+k_2-1)} & \text{if} \quad k_1 + k_2 \geq 1 \\ 0 & \text{if} \quad k_1 + k_2 = 0 \end{cases}, \qquad \mathfrak{a}(n) = \begin{cases} n^\alpha & \text{if } n \geq 1 \\ 1 & \text{if } n = 0 \end{cases}.$$



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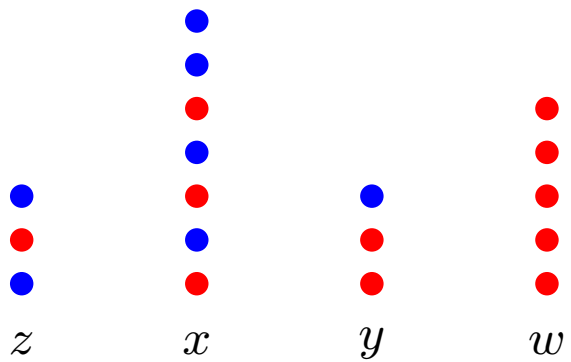
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Condensation

- **Stationary distribution**

$$\nu_N \propto \prod_{x\in S} \binom{|\boldsymbol{\eta}(x)|}{\eta_1(x)} \frac{m_1(x)^{\eta_1(x)} m_2(x)^{\eta_2(x)}}{\mathfrak{a}(|\boldsymbol{\eta}(x)|)}, \quad |\boldsymbol{\eta}(x)| = \eta_1(x) + \eta_2(x).$$

- **Regime**

α	Regime
$0 < \alpha < 1$	Sub-critical
$\alpha = 1$	Critical
$\alpha > 1$	Super-critical

- **Metastable well** For each $x \in S$ and $1 \ll \ell_N \ll N$,

$$\mathcal{E}_N^x := \{\boldsymbol{\eta} : |\boldsymbol{\eta}(x)| \geq N - \ell_N\}, \quad \mathcal{E}_N := \bigcup_{x\in S_\star} \mathcal{E}_N^x$$

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Theorem (Kim–L. arXiv : 2609.03821)

1. *For $\alpha \in (1, \infty)$, we have*

$$\lim_{N \to \infty} \nu_N(\mathcal{E}_N^x) = \frac{1}{|S_\star|} \quad (x \in S_\star) \quad \text{and} \quad \lim_{N \to \infty} \nu_N(\Delta_N) = 0. \tag{1}$$

2. *If $\alpha = 1$, condensation also occurs if ℓ_N additionally satisfies*

$$\lim_{N \to \infty} \frac{\log \ell_N}{\log N} = 1.$$

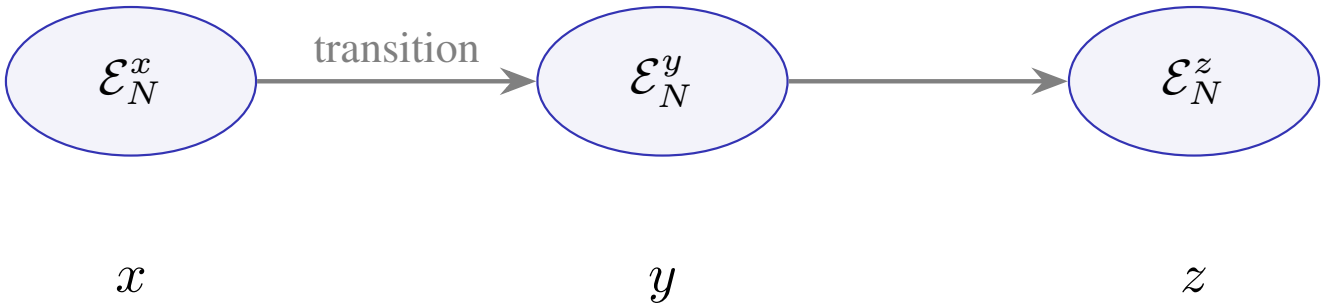
3. *If $\alpha \in (0, 1)$, there is no condensation phenomena, i.e.,*

$$\lim_{N \to \infty} \nu_N(\mathcal{E}_N^x) = 0, \quad \text{for all } x \in S.$$

Markov chain reduction

- **Trace process η_N^{tr}** Obtained by eliminating time spent in Δ_N.
- **Order process X_N**

$$\Psi_N = \sum_{x \in S_\star} x \mathbb{1}_{\mathcal{E}_N^x}, \quad X_N(t) = \Psi_N(\eta_N^{\mathrm{tr}}(t)).$$



Main theorem

- **Metastable time scale** $\theta_N = N^{\alpha+1}$.
- **Limiting Markov chain** $\mathbb{X}$ **on** $S_\star$

$$\mathfrak{r}(x,y) = \frac{1}{\Gamma(\alpha)\mathcal{I}_\alpha}\left(\frac{\rho}{\mathrm{cap}_1(x,y)} + \frac{1-\rho}{\mathrm{cap}_2(x,y)}\right)^{-1}, \quad x,y \in S_\star.$$

Theorem (Kim–L. arXiv : 2609.03821)

For sufficiently small $\gamma > 0$, *let* $\ell_N = \lfloor \gamma \log N \rfloor$. *Then for* $\alpha > 1$ *and for all* $x \in S_\star$,

1. ***(M) Local metastability*** *For* $b \in (0,1)$

$$\lim_{N\to\infty} \inf_{\boldsymbol{\eta}\in\mathcal{E}_N^x} \mathbb{P}_{\boldsymbol{\eta}}^N \left[H_{\boldsymbol{\xi}_N^x} < H_{\Omega_N \setminus \tilde{\mathcal{E}}_N^x} \right] = 1, \quad \tilde{\mathcal{E}}_N^x = \{\boldsymbol{\eta} \in \Omega_N : |\boldsymbol{\eta}(x)| \geq N - \lfloor N^b \rfloor\}$$

2. ***(C) Convergence*** $\{X_N(\theta_N t)\}_{t\geq 0} \xrightarrow[N\to\infty]{} \{\mathbb{X}(t)\}_{t\geq 0}$.
3. ***(D) Negligibility***

$$\lim_{N\to\infty} \sup_{\boldsymbol{\eta}\in\mathcal{E}_N^x} \mathbb{E}_{\boldsymbol{\eta}}^N \left[\int_0^T \mathbb{1}\{\boldsymbol{\eta}_N(\theta_N t) \in \Delta_N\} dt \right] = 0, \quad \text{for all} \quad T > 0.$$

Resolvent Approach

- **Resolvent Approach**: $\forall\, \mathbf{g} : S_\star \to \mathbb{R}$, $\lambda > 0$,

$$(\lambda - \theta_N \mathcal{L}_N) F_N = \sum_{x \in S_\star} \mathbf{g}(x) \mathbb{1}_{\mathcal{E}_N^x}, \quad (\lambda - \mathfrak{L}_{\mathbb{X}}) \mathfrak{f} = \mathbf{g},$$

$$\mathfrak{R}_{\mathfrak{L}_{\mathbb{X}}} : F_N \simeq \mathfrak{f}(x) \quad \text{on each } \mathcal{E}_N^x \ (x \in S_\star) \implies \textbf{(C)+(D)}$$

Landim–Marcondes–Seo, *J. Eur. Math. Soc.* '25

- $\mathfrak{R}_{\mathfrak{L}_{\mathbb{X}}} \iff \mathfrak{R}^{(1)} + \mathfrak{R}^{(2)}$:

  ⋆ $\mathfrak{R}^{(1)}$: F_N is asymptotically constant within each well $\mathcal{E}_N^x$:

$$\lim_{N \to \infty} \sup_{\boldsymbol{\eta}, \boldsymbol{\xi} \in \mathcal{E}_N^x} |F_N(\boldsymbol{\eta}) - F_N(\boldsymbol{\xi})| = 0, \quad \forall\, x \in S_\star.$$

  ⋆ $\mathfrak{R}^{(2)}$: Let $f_N(x) = \mathbb{E}_{\nu_N}[F_N \mid \mathcal{E}_N^x]$, then $f_N(x) \to \mathfrak{f}(x)$.

$\mathfrak{R}^{(1)}$ condition

Let $\boldsymbol{\xi}_N^x = A\delta_x^1 + B\delta_x^2$. It suffices to show that

$$\lim_{N\to\infty} \sup_{\boldsymbol{\eta}\in\mathcal{E}_N^x} |F_N(\boldsymbol{\eta}) - F_N(\boldsymbol{\xi}_N^x)| = 0, \quad \text{for each } x \in S_\star.$$

Let $H_{\boldsymbol{\xi}_N^x}$ be the $\boldsymbol{\xi}_N^x$-hitting time.

$$F_N(\boldsymbol{\eta}) = \mathbb{E}_{\boldsymbol{\eta}}^N \left[\int_0^\infty e^{-\lambda t} G_N(\boldsymbol{\eta}(\theta_N t)) dt \right], \quad G_N = \sum_{x\in S_\star} \mathbf{g}(x) \mathbb{1}_{\mathcal{E}_N^x}.$$

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Let $H_{\boldsymbol{\xi}_N^x}$ be the $\boldsymbol{\xi}_N^x$-hitting time. With change of variables and strong Markov property:

$$|F_N(\boldsymbol{\eta}) - F_N(\boldsymbol{\xi}_N^x)| \le 2\theta_N^{-1}\|\mathbf{g}\|_\infty \mathbb{E}_{\boldsymbol{\eta}}^N \left[H_{\boldsymbol{\xi}_N^x}\right]$$

Therefore, it suffices to show that

$$\sup_{\boldsymbol{\eta}\in\mathcal{E}_N^x} \mathbb{E}_{\boldsymbol{\eta}}^N \left[H_{\boldsymbol{\xi}_N^x}\right] \ll \theta_N$$

Construct a canonical path:

$$\boldsymbol{\eta} = \boldsymbol{\eta}_0, \boldsymbol{\eta}_1, \ldots, \boldsymbol{\eta}_k = \boldsymbol{\xi}_N^x, \quad k \le \kappa\ell_N = O(\log N).$$

$\mathfrak{R}^{(2)}$: idea

Key idea for $\mathfrak{R}^{(2)}$: Test the resolvent equation against $\mathfrak{H}_x^* := \mathfrak{H}^*_{\mathcal{E}_N^x, \mathcal{E}_N \setminus \mathcal{E}_N^x}$

$$\begin{cases} \mathcal{L}_N^* \mathfrak{H}_x^* \equiv 0 & \text{on} \quad \Omega_N \setminus \mathcal{E}_N \\ \mathfrak{H}_x^* \equiv 1 & \text{on} \quad \mathcal{E}_N^x \\ \mathfrak{H}_x^* \equiv 0 & \text{on} \quad \mathcal{E}_N \setminus \mathcal{E}_N^x \end{cases} .$$

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$$\langle (\lambda - \theta_N \mathcal{L}_N) F_N, \mathfrak{H}_x^* \rangle_{\nu_N} = \langle G_N, \mathfrak{H}_x^* \rangle_{\nu_N}$$

Definition of $\mathfrak{H}_x^*$ + condensation:

$$\lambda \langle F_N, \mathfrak{H}_x^* \rangle_{\nu_N} = \lambda f_N(x) \nu_N(\mathcal{E}_N^x) + o(1), \quad \text{and} \quad \langle G_N, \mathfrak{H}_x^* \rangle_{\nu_N} = \mathbf{g}(x) \nu_N(\mathcal{E}_N^x).$$

Only remains $\theta_N \langle -\mathcal{L}_N F_N, \mathfrak{H}_x^* \rangle_{\nu_N}$

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\theta_N \langle F_N, -\mathcal{L}_N^* \mathfrak{H}_x^* \rangle_{\nu_N} &= \theta_N \sum_{\boldsymbol{\eta} \in \mathcal{E}_N} F_N(\boldsymbol{\eta}) (-\mathcal{L}_N^* \mathfrak{H}_x^*)(\boldsymbol{\eta}) \nu_N(\boldsymbol{\eta}) \\
&\overset{\mathfrak{R}^{(1)}}{\simeq} \theta_N f_N(x) \sum_{\boldsymbol{\eta} \in \mathcal{E}_N^x} (-\mathcal{L}_N^* \mathfrak{H}_x^*)(\boldsymbol{\eta}) \nu_N(\boldsymbol{\eta}) \\
&\quad + \theta_N \sum_{y \in S_\star \setminus \{x\}} f_N(y) \sum_{\boldsymbol{\eta} \in \mathcal{E}_N^y} (-\mathcal{L}_N^* \mathfrak{H}_x^*)(\boldsymbol{\eta}) \nu_N(\boldsymbol{\eta})
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&\simeq -\nu_N(\mathcal{E}_N^x) \mathfrak{L}_{\mathbb{X}} f_N(x)
\end{aligned}$$

Then,

$$(\lambda - \mathfrak{L}_{\mathbb{X}}) f_N(x) \simeq \mathbf{g}(x) \ \ (x \in S_\star) \implies (\lambda - \mathfrak{L}_{\mathbb{X}})(f_N - \mathfrak{f}) \simeq 0.$$

However, $\mathfrak{H}_x^*$ is **unknown.**

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&\simeq -\nu_N(\mathcal{E}_N^x) \mathfrak{L}_{\mathbb{X}} f_N(x)
\end{aligned}$$

Approximate $\mathfrak{H}_x^*$ by $\mathbb{W}_x$:

$$\mathbb{W}_x|_{\mathcal{E}_N^x} \equiv 1, \quad \mathbb{W}_x|_{\mathcal{E}_N \setminus \mathcal{E}_N^x} \equiv 0, \quad \theta_N \langle -\mathcal{L}_N F_N, \mathfrak{H}_x^* \rangle_{\nu_N} \simeq \theta_N \langle -\mathcal{L}_N F_N, \mathbb{W}_x \rangle_{\nu_N}$$

H^1-approximation

- **Uniform bound**

$$\sup_{N\geq 1} \theta_N D_N(F_N) < \infty$$

- **Sector condition** $\langle G, -\mathcal{L}_N F\rangle_{\nu_N}^2 \lesssim D_N(G) D_N(F)$.

- Together,

$$|\theta_N \langle F_N, -\mathcal{L}_N^*(\mathfrak{H}_x^* - \mathbb{W}_x)\rangle_{\nu_N}|^2 \lesssim \theta_N^2 D_N(F_N) D_N(\mathfrak{H}_x^* - \mathbb{W}_x).$$

- Approximate $\mathbb{W}_x \approx \mathfrak{H}_x^*$

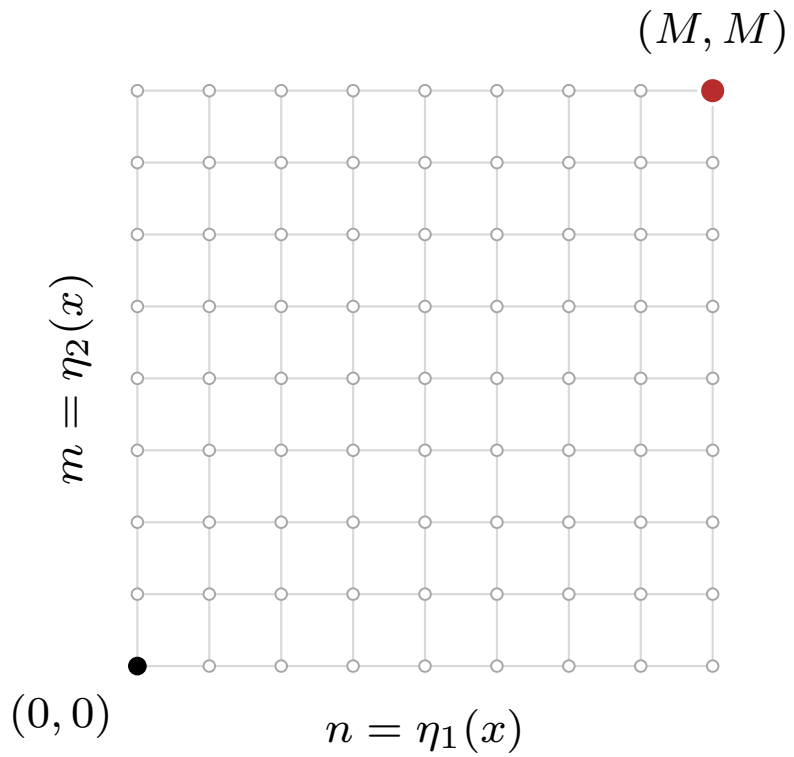
$$\theta_N D_N(\mathfrak{H}_x^* - \mathbb{W}_x) = o(1),$$

(in H^1 sense, Kim–Seo *Ann. Probab.* '25), we can replace $\mathfrak{H}_x^*$ with $\mathbb{W}_x$!

$\mathfrak{R}^{(2)}$: Reduced model

Two sites $S = \{x, y\}$, M particles of each species and $N = 2M$.

$$(n, m) = (\eta_1(x), \eta_2(x)), \quad \eta(y) = (M - n, M - m), \quad \mathcal{E}_N^z = \{\boldsymbol{\xi}_N^z\}, \ (z \in \{x, y\}).$$



Two condensates

Both species are at x:

$$(n, m) = (M, M).$$

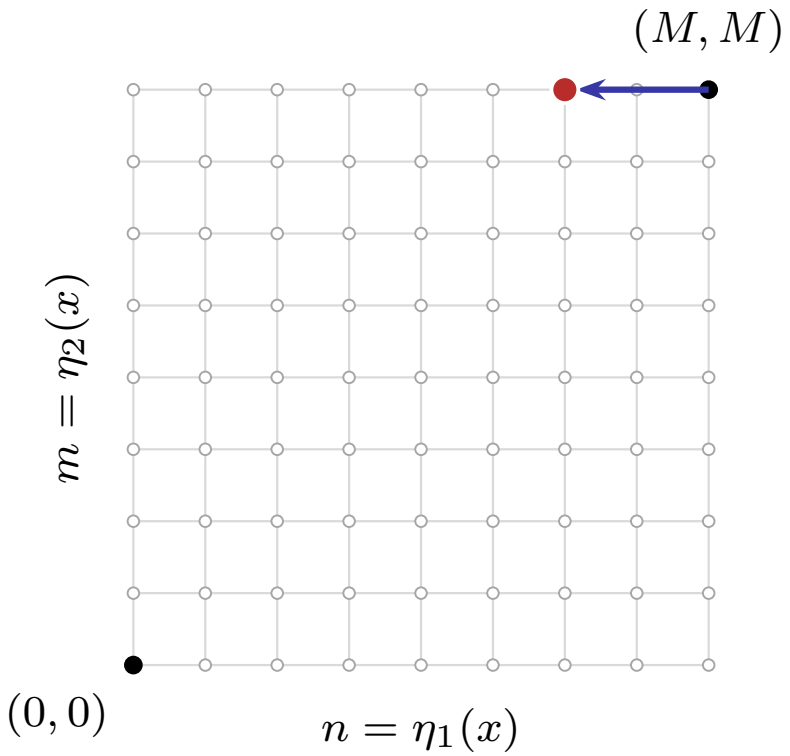
The other corner $(0, 0)$ represents condensation at y.

We seek a test function equal to 1 and 0 at these corners.

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A single-particle jump

One species-1 particle moves from x to y:

$$(M, M) \longrightarrow (M - 1, M).$$

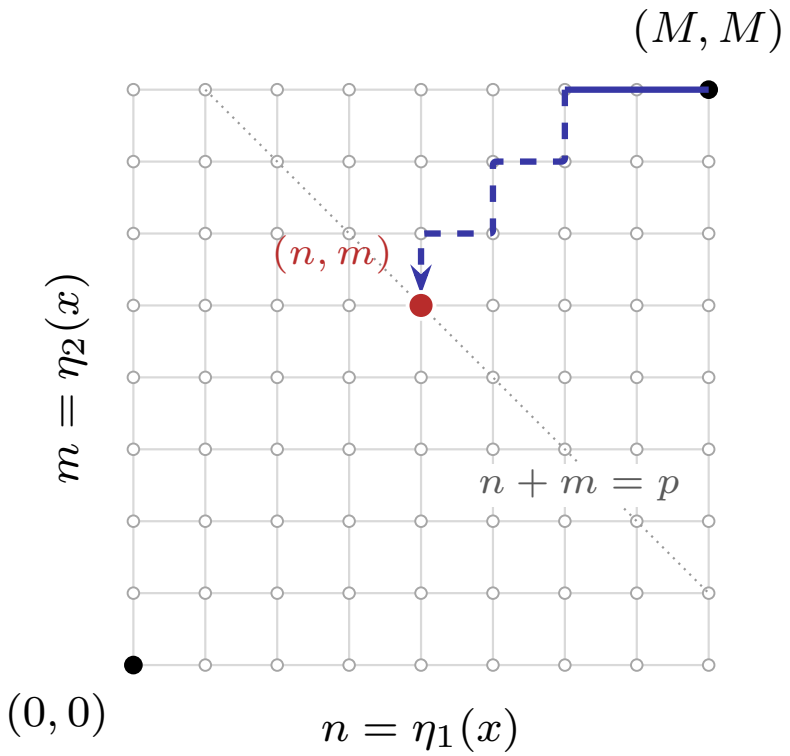
Its jump rate is

$$\tfrac{1}{2} g(2M)\, r_1(x, y).$$

$\mathfrak{R}^{(2)}$: Reduced model

Two sites $S = \{x, y\}$, M particles of each species and $N = 2M$.

$$(n, m) = (\eta_1(x), \eta_2(x)), \quad \eta(y) = (M - n, M - m), \quad \mathcal{E}_N^z = \{\boldsymbol{\xi}_N^z\}, \ (z \in \{x, y\}).$$



An intermediate configuration

Stop at a state with

$$p = n + m$$

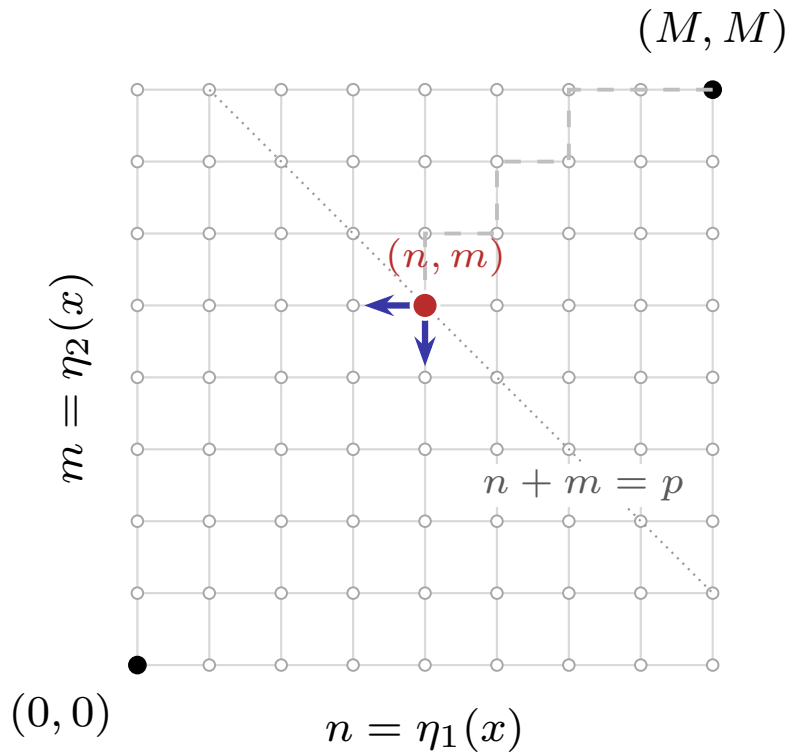
$$\eta(x) = (n, m), \quad \eta(y) = (M-n, M-m).$$

The dashed segment is an illustrative sequence of $x \to y$ jumps.

$\mathfrak{R}^{(2)}$: Reduced model

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Two ways to transfer one particle

For the $x \to y$ jumps at $p = n + m$,

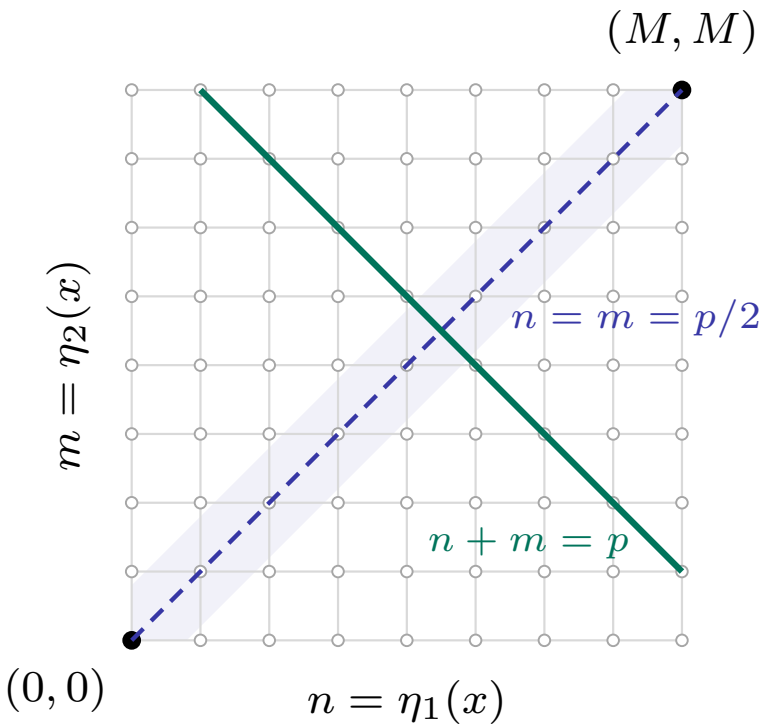
$$q_{\leftarrow} = \frac{n}{p} \cdot \frac{p^{\alpha}}{(p-1)^{\alpha}} r_1(x, y),$$

$$q_{\downarrow} = \frac{m}{p} \cdot \frac{p^{\alpha}}{(p-1)^{\alpha}} r_2(x, y).$$

Both lower p by one, but change $n - m$ by -1 and $+1$, respectively.

Reduced model: why a composition corrector?

$$p = n + m, \qquad \rho = \tfrac{1}{2}, \qquad R_i = r_i(x, y) + r_i(y, x).$$



A total-mass profile

The one-dimensional energy estimate suggests

$$H_p = \frac{\sum_{k=0}^{p-1} a(k)a(N-k-1)}{\sum_{k=0}^{N-1} a(k)a(N-k-1)}.$$

Thus $H_0 = 0$ and $H_N = 1$.

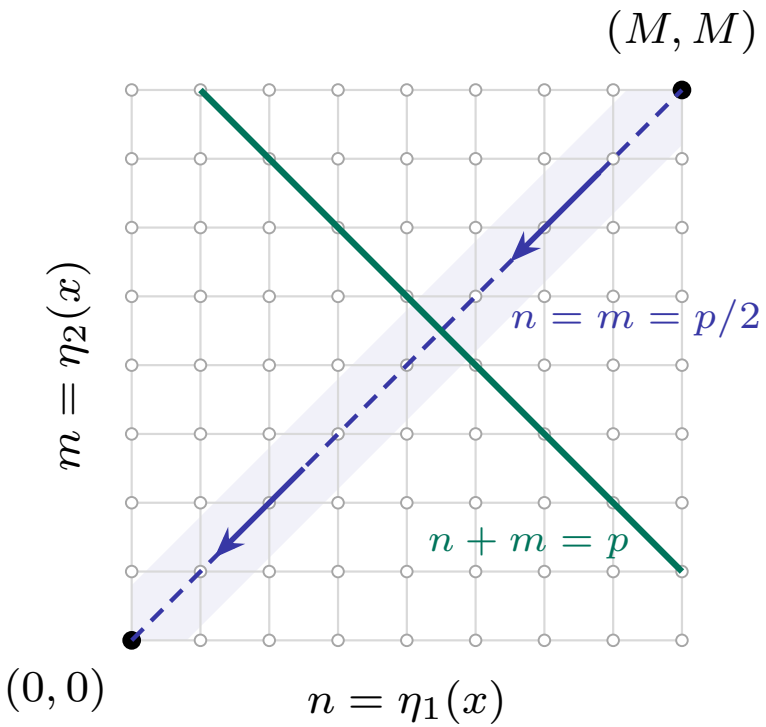
If $R_1 \neq R_2$, optimal increments in the two species directions differ:

$$\nabla_1 G : \nabla_2 G \approx R_1^{-1} : R_2^{-1}.$$

A function of p alone has equal increments in both directions.

Reduced model: why a composition corrector?

$$p = n + m, \qquad \rho = \tfrac{1}{2}, \qquad R_i = r_i(x, y) + r_i(y, x).$$



Conditional composition

Under the stationary law, given $n + m = p$,

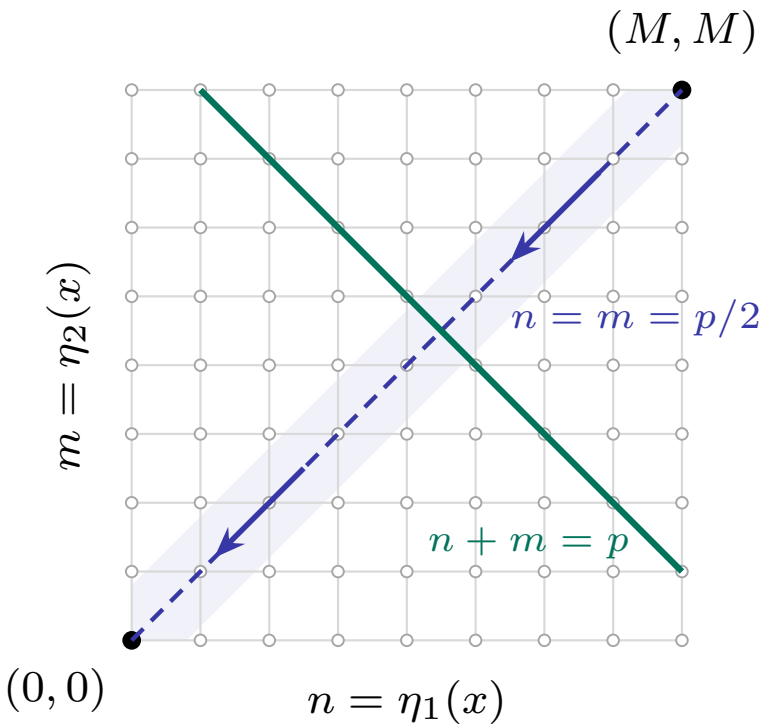
$$\nu_N(n, p - n \mid p) = \frac{\binom{p}{n}\binom{N-p}{M-n}}{\binom{N}{M}}.$$

Deviation

$$n - \frac{p}{2} = \frac{n - m}{2}, \quad \mathbb{E}[n \mid p] = \frac{p}{2}.$$

Reduced model: why a composition corrector?

$$p = n + m, \qquad \rho = \tfrac{1}{2}, \qquad R_i = r_i(x, y) + r_i(y, x).$$



Correcting for unequal rates

$$\boxed{G(n, m) = H_p + \beta_p \left(n - \frac{p}{2} \right)}$$

$$\beta_p = \frac{H_p - H_{p-1}}{\vartheta} \left(\frac{1}{R_1} - \frac{1}{R_2} \right), \quad \beta_0 = 0,$$

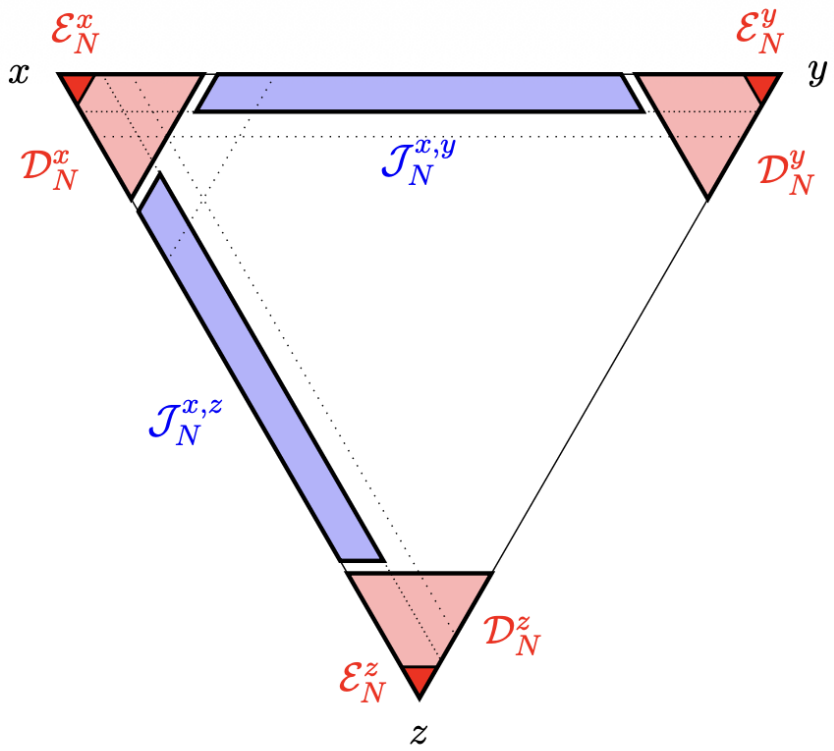
$$\vartheta = \frac{1}{2R_1} + \frac{1}{2R_2}.$$

The correction has zero conditional mean and adjusts the two directional increments.

This is the prototype for the test function on the tube.

Wells, tubes, and the global test function

Choose $\pi_N = \lfloor N^\beta \rfloor$ where $\beta \in (\frac{1}{\alpha}, 1)$.



- **Enlarged wells**

$$\mathcal{D}_N^z := \{\boldsymbol{\eta} : |\boldsymbol{\eta}(z)| \geq N - \pi_N\}, \ z \in S_\star$$

- **Tubes** For $x \neq y \in S_\star$

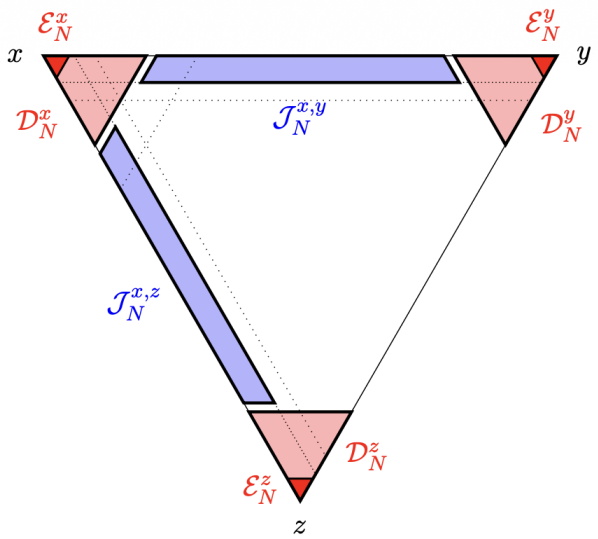
$$\mathcal{T}_N^{x,y} = \{\boldsymbol{\eta} : |\boldsymbol{\eta}(x)| + |\boldsymbol{\eta}(y)| \geq N - \pi_N\},$$

$$\mathcal{J}_N^{x,y} = \mathcal{T}_N^{x,y} \setminus (\mathcal{D}_N^x \cup \mathcal{D}_N^y).$$

- **Good region**

$$\mathcal{G}_N^x := \bigcup_{z \in S_\star} \mathcal{D}_N^z \cup \bigcup_{y \in S_\star \setminus \{x\}} \mathcal{J}_N^{x,y}$$

Wells, tubes, and the global test function



- **Enlarged wells**

$$\mathcal{D}_N^z := \{\boldsymbol{\eta} : |\boldsymbol{\eta}(z)| \geq N - \pi_N\},\ z \in S_\star$$

- **Tubes** For $x \neq y \in S_\star$

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- **Good region**

$$\mathcal{G}_N^x := \bigcup_{z \in S_\star} \mathcal{D}_N^z \cup \bigcup_{y \in S_\star \setminus \{x\}} \mathcal{J}_N^{x,y}$$

Global test function

For each $x \in S_\star$,

$$\mathbb{W}_x(\boldsymbol{\eta}) = \sum_{y \in S_\star \setminus \{x\}} \Theta_\epsilon^{x,y}(\boldsymbol{\eta}/N)\, \mathcal{W}_{x,y}(\boldsymbol{\eta}),$$

$$\mathbb{W}_x\big|_{\mathcal{E}_N^x} \equiv 1, \qquad \mathbb{W}_x\big|_{\mathcal{E}_N \setminus \mathcal{E}_N^x} \equiv 0, \qquad \mathbb{W}_x\big|_{\mathcal{J}_N^{x,y}} = \mathcal{W}_{x,y}.$$

$\mathfrak{R}^{(2)}$: Testing with $\mathbb{W}_x$

$$\langle (\lambda - \theta_N \mathcal{L}_N) F_N, \mathbb{W}_x \rangle_{\nu_N} = \langle G_N, \mathbb{W}_x \rangle_{\nu_N}.$$

- $\mathbb{W}_x$ definition + condensation

$$\lambda f_N(x) \nu_N(\mathcal{E}_N^x) + \theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N} = \mathbf{g}(x) \nu_N(\mathcal{E}_N^x) + o(1)$$

- Only remains $\theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N}$,

$$\theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N} = \theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N, \mathcal{G}_N^x} + \theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N, (\mathcal{G}_N^x)^c}.$$

- **Motivation** For $\mathcal{E} \subset \Omega_N$, $\langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N, \mathcal{E}}$

$$\sum_{i \in \{1,2\}} \sum_{\boldsymbol{\eta} \in \mathcal{E}} \sum_{\substack{w \in S \\ u : \eta_i(u) \geq 1}} \nu_N(\boldsymbol{\eta}) \boldsymbol{g}_i(\boldsymbol{\eta}(u)) r_i^*(u, w) F_N(\boldsymbol{\eta}) [\mathbb{W}_x(\boldsymbol{\eta}) - \mathbb{W}_x(\boldsymbol{\eta} - \delta_u^i + \delta_w^i)]$$

change $\boldsymbol{\eta} - \delta_u^i \to \boldsymbol{\xi} \in \Omega_{N-1}^{i,-}$

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$$\langle (\lambda - \theta_N \mathcal{L}_N) F_N, \mathbb{W}_x \rangle_{\nu_N} = \langle G_N, \mathbb{W}_x \rangle_{\nu_N}.$$

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- **Motivation** For $\mathcal{E} \subset \Omega_N$, $\langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N, \mathcal{E}}$

$$\sum_{\boldsymbol{\xi} \in \Omega_{N-1}^{i,-}} \sum_{\substack{u,w \in S \\ \boldsymbol{\xi} + \delta_u^i \in \mathcal{E}}} \frac{\boldsymbol{m}^{\boldsymbol{\xi}}}{\boldsymbol{g}!(\boldsymbol{\xi})} m_i(u) r_i^*(u,w) F_N(\boldsymbol{\xi} + \delta_u^i) [\mathbb{W}_x(\boldsymbol{\xi} + \delta_u^i) - \mathbb{W}_x(\boldsymbol{\xi} + \delta_w^i)]$$

$\mathfrak{R}^{(2)}$: Testing with $\mathbb{W}_x$

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$$\sum_{i \in \{1,2\}} \sum_{\boldsymbol{\xi}} \frac{\boldsymbol{m}^{\boldsymbol{\xi}}}{\boldsymbol{g}!(\boldsymbol{\xi})} \sum_{u,w \in S} m_i(u) r_i^*(u,w) [\mathbb{W}_x(\boldsymbol{\xi} + \delta_u^i) - \mathbb{W}_x(\boldsymbol{\xi} + \delta_w^i)] = 0$$

$\mathfrak{R}^{(2)}$: Testing with $\mathbb{W}_x$

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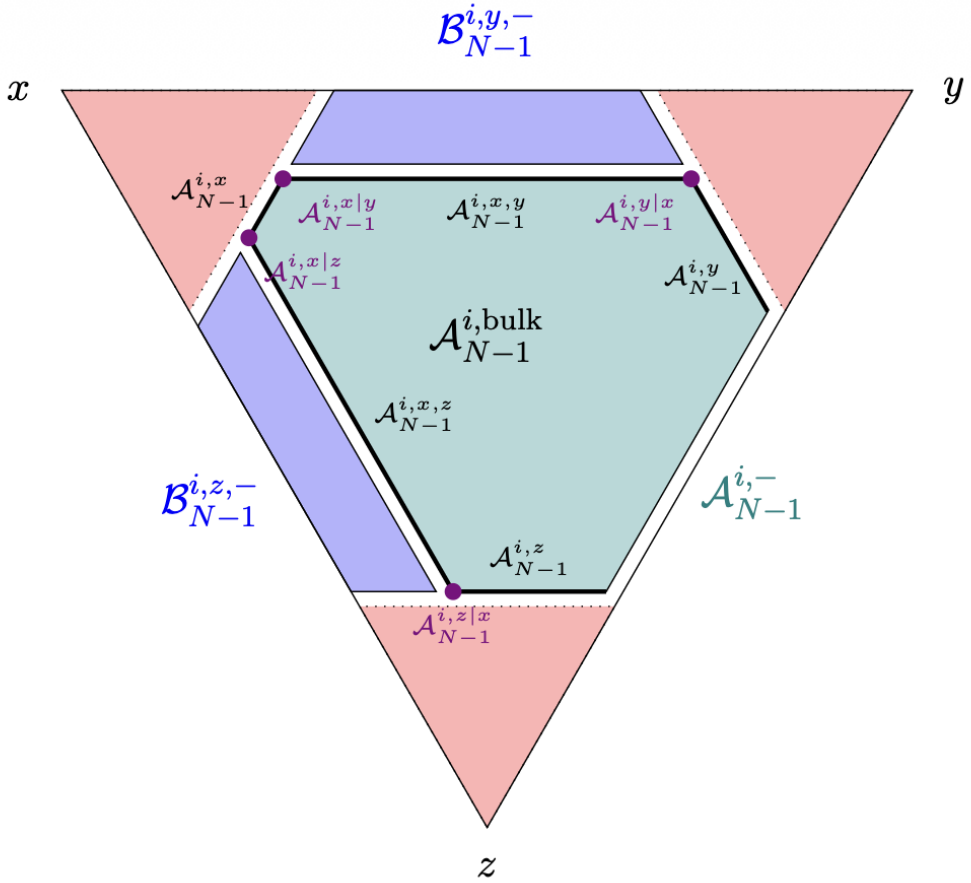
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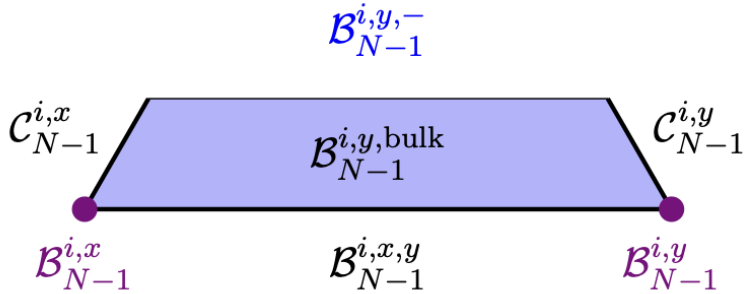
$$\theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N} = \theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N, \mathcal{G}_N^x} + \theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N, (\mathcal{G}_N^x)^c}.$$

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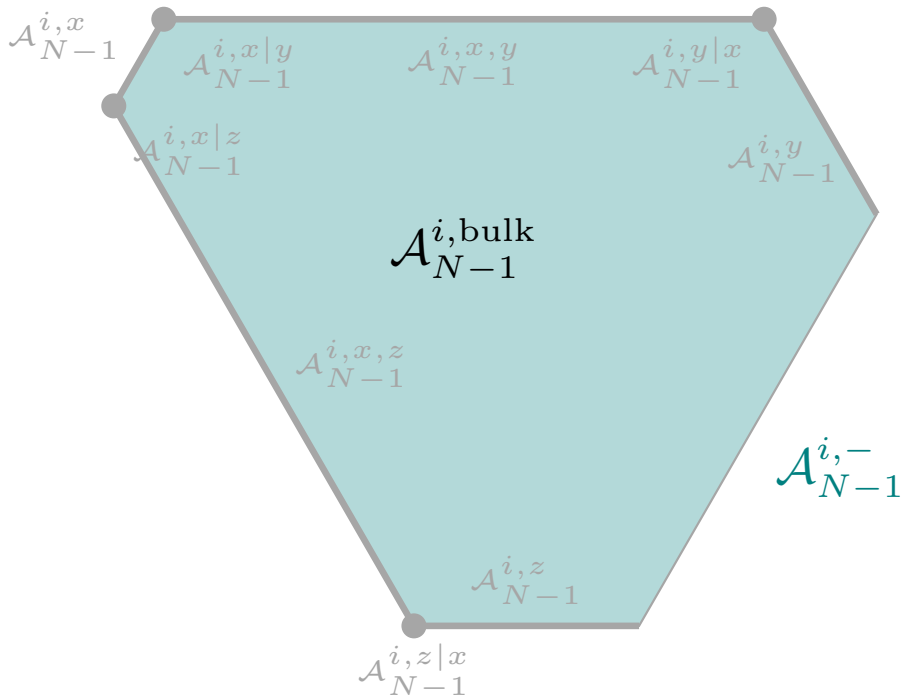
$$\sum_{i \in \{1,2\}} \sum_{\boldsymbol{\xi}} \frac{\boldsymbol{m}^{\boldsymbol{\xi}}}{\boldsymbol{g}!(\boldsymbol{\xi})} \sum_{u,w \in S} m_i(u) r_i^*(u,w) \left(F_N(\boldsymbol{\xi} + \delta_u^i) - \frac{1}{|S|} \sum_{u \in S} F_N(\boldsymbol{\xi} + \delta_u^i) \right) \times [\mathbb{W}_x(\boldsymbol{\xi} + \delta_u^i) - \mathbb{W}_x(\boldsymbol{\xi} + \delta_w^i)]$$

$N-1$ decomposition





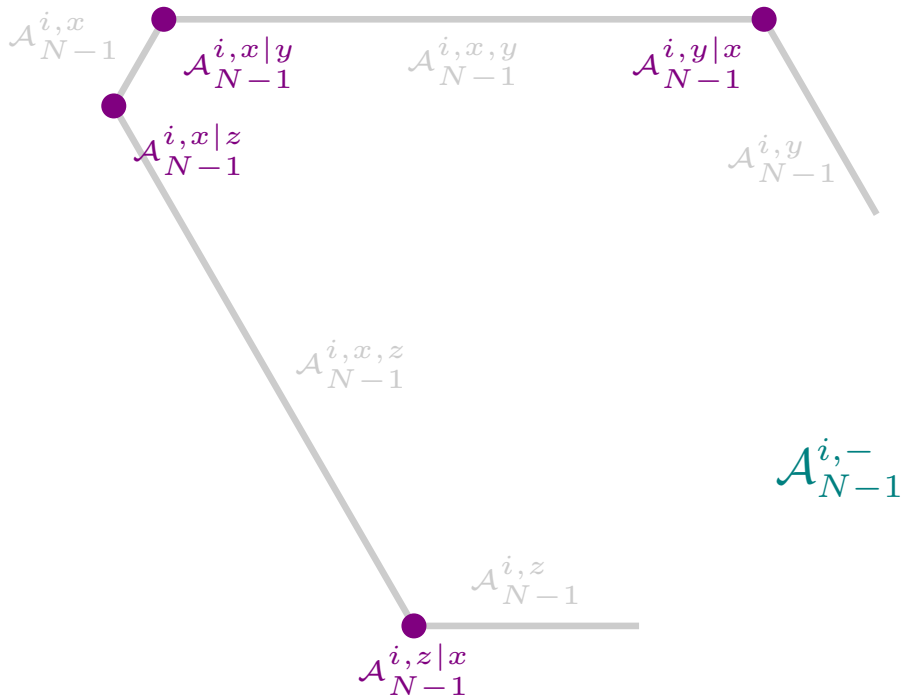
$N-1$ decomposition



Outside bulk Centering and Cauchy–Schwarz give

$$|\text{outside bulk term}| \lesssim \underbrace{\sqrt{\theta_N D_N(F_N)}}_{O(1)} \underbrace{\sqrt{\theta_N D_N(\mathbb{W}_x; (\mathcal{G}_N^x)^c)}}_{o_N(1)}.$$

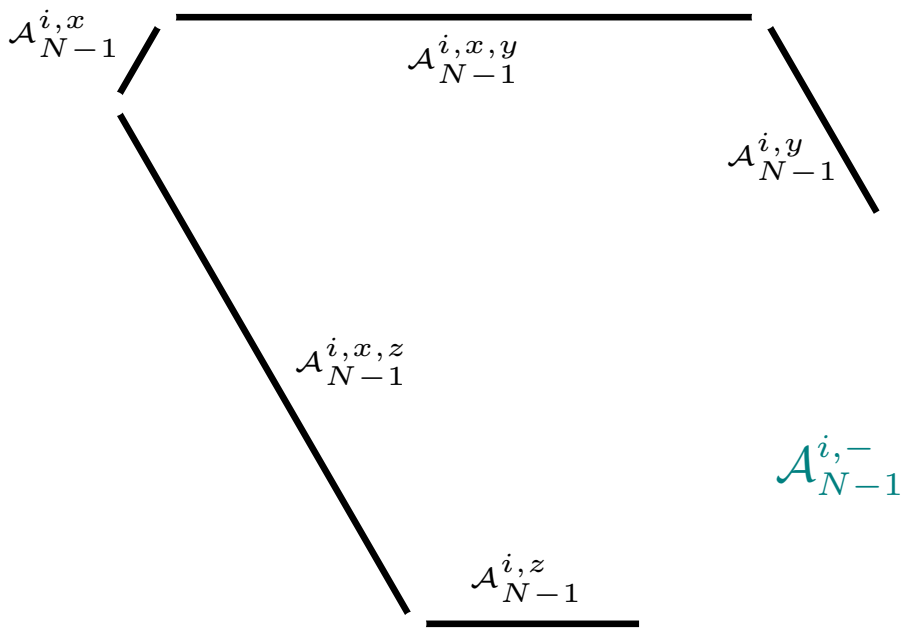
$N-1$ decomposition



Edge adjacent to wells Since $\mathbb{W}_x$ is constant

$$\text{contribution} = 0$$

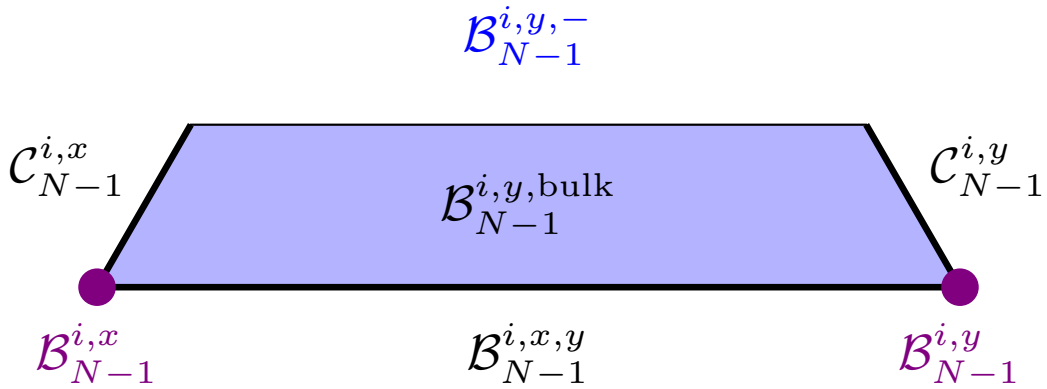
$N-1$ decomposition



Remaining edge For fixed ϵ,

$$|\text{remaining edge}| \;\le\; C_\epsilon \frac{N}{\pi_N^\alpha} \;=\; o_N(1), \qquad N \ll \pi_N^\alpha.$$

$N-1$ decomposition



Region	Contribution	Main reason
Edge adjacent to wells	0	$\mathbb{W}_x$ is constant
Remaining edge	$o_N(1)$	Small stationary mass
Tube bulk $\mathcal{B}_{N-1}^{i,y,\mathrm{bulk}}$	$\nu_N(\mathcal{E}_N^x)\mathfrak{r}(x,y)[f_N(x)-f_N(y)]+o(1)$	Conditional averaging

Completing $\mathfrak{R}^{(2)}$

Summing all contributions,

$$\begin{aligned}\theta_N \langle F_N, -\mathcal{L}_N^* \mathbb{W}_x \rangle_{\nu_N} &= \nu_N(\mathcal{E}_N^x) \sum_{y \neq x} \mathfrak{r}(x,y)[f_N(x) - f_N(y)] + o(1) \\ &= -\nu_N(\mathcal{E}_N^x) \mathfrak{L}_{\mathbb{X}} f_N(x) + o(1).\end{aligned}$$

Therefore, resolvent equation becomes

$$\lambda f_N(x) \nu_N(\mathcal{E}_N^x) - \nu_N(\mathcal{E}_N^x) \mathfrak{L}_{\mathbb{X}} f_N(x) = \mathbf{g}(x) \nu_N(\mathcal{E}_N^x) + o(1).$$

Since $\nu_N(\mathcal{E}_N^x) \to |S_\star|^{-1}$,

$$(\lambda - \mathfrak{L}_{\mathbb{X}}) f_N(x) = \mathbf{g}(x) + o(1), \quad \forall x \in S_\star.$$

The $\mathfrak{R}^{(2)}$ condition

Let $(\lambda - \mathfrak{L}_{\mathbb{X}})\mathfrak{f} = \mathbf{g}$, then we obtain

$$\|f_N - \mathfrak{f}\|_\infty \xrightarrow[N \to \infty,\ \epsilon \downarrow 0]{} 0.$$

Further topics

Q. Generalize for super-critical k-species case?

With time scale $\theta_N = N^{\alpha+1}$,

$$\mathfrak{r}(x,y) = \frac{1}{\Gamma(\alpha)\mathcal{I}_\alpha} \left(\sum_{j=1}^{k} \frac{\rho_j}{\mathrm{cap}_j(x,y)} \right)^{-1}$$

Q. Can we obtain analogous results at $\alpha = 1$?

The difficult part is the local metastability for larger wells.

Thank you!